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RESEARCH ARTICLE

UFTrans: A Hybrid 3D Vision Transformer and U-Net Model for Brain Tumor Segmentation

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ABSTRACT Gliomas represent a highly common and malignant form of brain tumor. Precise segmentation plays a pivotal role in supporting clinical diagnosis and guiding treatment strategies. However, many existing techniques struggle to integrate both local and global information, which limits their segmentation accuracy. Local features such as tumor texture, boundaries, and subtle structural details are important to delineate subregions like the necrotic core and enhance tumor boundaries. Global features, such as tumor size, shape, and spatial relationships within the brain, provide crucial contextual information for precise localization. To address these challenges, we propose UFTrans, a hybrid model that effectively fuses the local feature extraction capabilities of the 3D attention U-Net with the global contextual understanding of a transformer-based model. Our approach employs Hadamard product-based feature fusion to selectively emphasize important features and suppress redundant information, creating a composite feature representation. An inherent attention mechanism in the 3D U-Net continues to refine segmentation by eliminating unwanted activations to reduce computational complexity and accelerate model convergence. We further employ tumor-centered patching to address the severe class imbalance in brain tumor datasets. Extensive experiments on the BraTS 2021 dataset demonstrate the model's effectiveness, achieving Dice scores of 0.90, 0.91, and 0.89 for the whole tumor, enhancing tumor and tumor core regions, respectively, on the validation set. Comprehensive ablation studies validate the contribution of each architectural component. These results surpass state-of-the-art approaches, indicating the ability of the model to enhance brain tumor segmentation and clinical decision making.

INDEX TERMS Medical image segmentation, brain tumor, vision transformer, 3D attention U-Net, deep learning.

I. INTRODUCTION

The Global Cancer Statistics 2020 report estimated 308,102 new cases and 251,329 deaths due to brain tumors around the world [1]. Primary brain tumors are the 10th most common reason for death related to cancer deaths. Gliomas account for 30–40% of brain tumors and 81% of malignant cases in adults and are a significant contributor to brain tumor-related deaths. Surgical resection is considered the most effective treatment that focuses on the critical need for accurate and automated tumor segmentation in clinical evaluation and diagnosis [2], [3]. Gliomas are a malignant form of primary brain tumor originating from abnormal glial cell proliferation in the cerebrum or cerebellum and pose a life-threatening risk if not diagnosed early and adequately. Glioma is categorized into high-grade glioma (HGG) and low-grade glioma (LGG) based on its degree of aggressiveness. LGGs usually grow slowly with a better prognosis, while HGGs grow fast and are aggressive. Distinguishing between these two types is vital for monitoring disease advancement and for devising appropriate treatment strategies [4].

Survival rates among glioma patients can differ based on tumor grade and the individual's age. Patient survival rates are closely associated with glioma type. The 5-year

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survival rate for low-grade gliomas can reach approximately 80%, whereas for high-grade gliomas it typically falls below 5% [5]. The prognosis is worse for the patients who are older during diagnosis and treatment. The recovery rate for these brain tumors is approximately 75% for individuals under 15 and 72% for people under the age group of 15 to 39 in the last five years and survival chances for people age 40 and older are 21%. In the United States, around 18,600 individuals lost their lives to primary cancerous brain and CNS tumors in 2021.

Preserving local features while ensuring efficient tumor segmentation remains a significant challenge in medical image analysis [6]. Moreover, accurate delineation of tumor sub-regions, including the necrotic core, edema, and enhancing regions, is essential for effective treatment planning using MRI scans. As a non-invasive modality, MRI provides high-resolution brain images without skull-induced artifacts, offering comprehensive insights for tumor evaluation [7]. Additionally it offers high soft tissue contrast and absence of electrical radiation. These scans provide vital information for precise diagnosis with diverse imaging modalities like contrast-enhanced T1ce, T1, T2, and Fluid Attenuation Inversion Recovery (FLAIR). Each modality offers unique information for brain tumor imaging [8].

T1-weighted images are outstanding for visualizing the brain's anatomy and typically tumors appear as dark areas on these images, but this may vary depending on the tumor type and composition. Furthermore, due to heightened water content edema tissues could display hypointense on T1-weighted images. T2-weighted are sensitive to areas with water content and are useful for locating edema (swelling) that often appear brighter on T2-weighted images due to increased water content. These images provide information about the extent of peritumoral edema and its relation with the tumor and surrounding brain tissue. T1 Contrast-enhanced MRI is particularly useful for identifying areas of active tumor growth, as tumors typically exhibit increased vascularity and accumulate contrast agents. FLAIR imaging is considered to suppress the signal from cerebrospinal fluid (CSF), making it suitable to detect abnormalities near CSF-filled spaces. FLAIR sequences are sensitive to tissue water content and are valuable for detecting peritumoral edema and tumor infiltration into surrounding brain tissue. Tumors typically appear hyperintense on FLAIR images, and the extent of FLAIR signal abnormality can help assess tumor burden and response to treatment. Combining these different MRI sequences allows radiologists and clinicians to comprehensively understand brain tumor regarding their location, size, morphology, vascularity, and relationship with surrounding structures. Three MRI modalities, Fluid Attenuation Inversion Recovery (FLAIR), T2, and T1-Contrast Enhanced are shown in Figure 1.

Many image processing and deep learning methods are used in research to build systems for tumor classification and segmentation. Segmentation technique detects tumor from pixels and is becoming a progressively popular topic in medical imaging. Tumors specifically glioma diagnosis greatly depend on medical image segmentation. Oligodendrogliomas and astrocytoma are both common and have minimum risk, but high-grade glioma is the most critical type of tumor it can cause more harm than low-grade glioma. Typically, gliomas are treated using chemotherapy, surgery, and radiotherapy. Early detection of glioma is essential and it can help in the treatment plan and diagnosis of patients.

Deep learning approaches used in brain tumor segmentation have demonstrated outstanding results and enable medical domain experts to provide enhanced healthcare facilities to the population. The emergence of attention mechanisms have enabled models to selectively focus on semantically critical features, improving performance in complex tasks such as question answering and medical image analysis [9]. Enhancements in deep learning, integrating artificial intelligence and machine learning approaches, have facilitated the creation of various cutting-edge techniques to detect tumors and allow the early identification of tumors. Convolutional Neural Networks (CNNs) have been extensively utilized in MRI-based segmentation, particularly for brain tumor analysis. Among these, U-Net and its variants demonstrate strong performance owing to their ability to preserve local features and derive discriminative information from MRI data [10]. Moreover, incorporating attention mechanisms enables the network to prioritize relevant regions and highlight essential characteristics throughout the segmentation process [11]. These technologies enable the implementation of protective measures that contribute to saving lives. Despite the remarkable progress made by deep learning models in medical image segmentation, many existing techniques, especially self-attention-based models, struggle to capture fine local features such as texture and blurred boundaries, often due to partial volume effects in glioma segmentation.

On the other hand, approaches that focus only on edge and boundary features excel in capturing subregion details but fail to capture global features like the overall shape, size, and location of tumors in the brain region. This lack of complete feature representation leads to suboptimal segmentation results as the models fail to find the fine balance between local (texture, edges, blurred boundaries) and global features (size and shape of tumor within brain). Local features provide critical information about the fine structure and boundaries of tumors. In contrast, global features are necessary to understand the overall tumor location, size, and shape within the brain region. Therefore, the key challenge in glioma segmentation lies in balancing local and global feature extraction. Hence, a hybrid approach is used to overcome these challenges and to achieve better segmentation results.

The overall contribution of work is given below:

1) A hybrid model is developed that integrates local details from the 3D attention U-Net encoder with long-range dependencies from the transformer via Hadamard product-based feature fusion, which

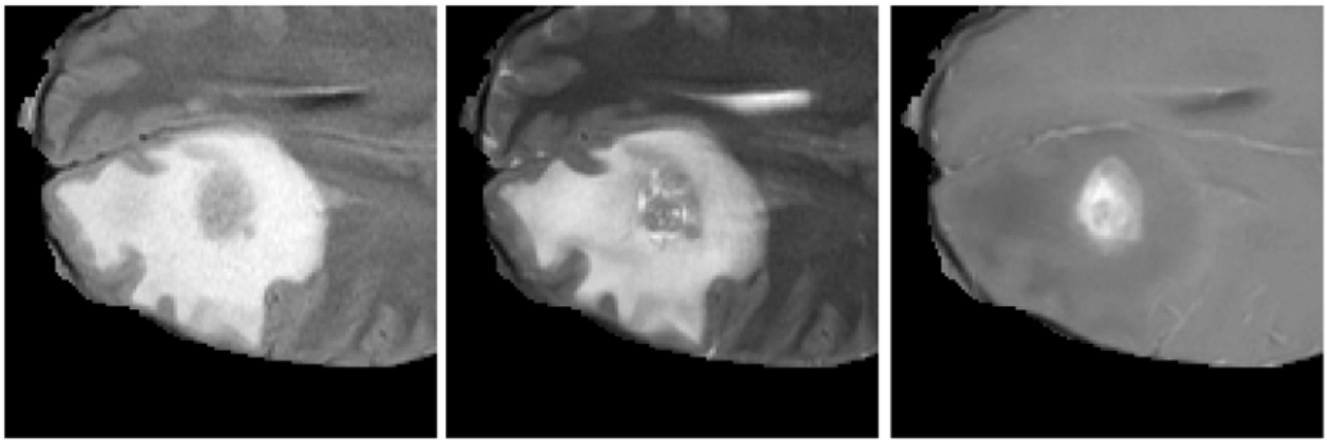


FIGURE 1. Brain tumor MRI Scans of Flair, T2, T1ce modalities in axial view from BraTS 2021 dataset.

suppresses insignificant features while accentuating essential ones.

2) To reduce computational overhead by minimizing irrelevant activations, we incorporate an attention mechanism in 3D U-Net. The attention weights derived from 3D U-Net optimizes the transmission of information via skip connections. This ensures that only the most significant features are kept, leading to faster model convergence.
3) Comprehensive ablation studies validating the contribution of each architectural component, including fusion strategies (Hadamard vs. concatenation vs. addition), attention mechanisms, and transformer integration.
4) State-of-the-art results on BraTS 2021 by considering a stack of three modalities (T2, T1ce, Flair) as input, achieving Dice scores of 0.90 (WT), 0.91 (ET), and 0.89 (TC).

Contribution of UFTrans is a targeted architectural advancement at the level of fusion placement (pre-decoder, multi-scale), fusion operation (parameter-free multiplicative gating via Hadamard product), and an attention-before-fusion pipeline that ensures only salient local features participate in the cross-branch interaction. These three coupled choices are validated by comprehensive ablation studies (Tables 8, 9, 10 and 11), which demonstrate systematic and consistent improvements across all tumor subregions.

The rest of this paper is organized as follows. Section II introduces the related work on brain tumor segmentation techniques. Section III discusses the proposed approach. Section IV describes the experimental setup and results. Section V highlights the limitations and future work. Section VI concludes the paper.

II. RELATED WORK

In the past few years, there has been an increasing emphasis on automated segmentation methods. Despite the challenges posed by traditional segmentation techniques, researchers continue to develop new approaches aimed at achieving accurate and reliable results. The section below highlights and categorizes recent methods showing different deep learning-based segmentation techniques.

A. HYBRID MODELS (CNN + TRANSFORMER-BASED ARCHITECTURES)

In a recent study, the authors have proposed a model called Swin UNETR that introduces a novel transformer-based architecture for segmenting 3D MRI volumes. Their model combines the strengths of the Swin Transformer for hierarchical self-attention and a U-Net like CNN-based decoder. Using the concept of multiscale feature extraction this model performs well without any extensive preprocessing and achieves competitive dice scores on BraTS 2021 dataset [12].

In another study, the authors developed a fully transformer-based 3D model for MRI volume segmentation, addressing some of the limitations in previous transformer-based models. In contrast to earlier techniques, nnFormer integrates convolution with self-attention and incorporates both local and global volume-based self-attention. It also replaces conventional skip connections with skip attention to optimize the transfer of information between the encoder and decoder. nnFormer demonstrated superior performance across three publicly available datasets, and the Hausdorff distance(HD95) is reduced. The HD95 score makes it suitable for accurate boundary delineation. nnFormer was combined with U-Net through a set of components, making it a highly effective model for medical image segmentation [13].

Similarly, the authors presented MedNeXt, a modernized convolution architecture specially designed for medical image segmentation inspired by recent advancements in Transformer-based models and ConvNeXt architecture. MedNeXt filled the gap between the high inductive biases of CNN and the long-range dependency modelling capabilities of the transformers model. The authors introduced a customized Encoder-Decoder network with Residual ConvNeXt up

sampling and down sampling block that maintains semantic richness across scales. MedNeXt iteratively increased kernel slices to address the risk of performance saturation on small datasets. Furthermore, to optimize performance, they incorporated compound scaling at multiple levels, adjusting depth, width, and kernel size [14].

The authors also introduced UNETR, a segmentation model that reformulates the segmentation task as a sequence modeling framework prediction issue by leveraging a transformer as the encoder. The proposed transformer encoder is able to effectively extract informative representations from the raw input data. A U-Net based architecture has been adopted, retaining the contracting-expanding path and integrating skip connections between the transformer encoder and decoder to optimize the information flow process for accurate semantic segmentation. The results obtained on BTCV and MSD databases have shown outstanding performance for brain tumor segmentation tasks. [15].

Another contribution put forward the MVSI-Net, which is a novel deep learning based model that utilizes the concept of multiview attention and multiscale feature interaction to improve the segmentation of brain tumors in MRI scans. As compared to conventional models based on U-Net architecture, MVSI-Net employs a multiview attention (MVA) structure both in the encoder and decoder to incorporate global as well as local context. Moreover, the use of multiscale feature interaction (MSI) module in the bottleneck region allows for capturing of the relevant features from various scales along with minimizing the redundancy. MVSI Net was validated on the BraTS datasets and yielded Dice scores of 0.85, 0.86, and 0.78 for whole tumor, enhancing tumor, and tumor core respectively. Thus, MVSI-Net performed competitively compared to other advanced models [16].

The next contribution introduced TransU2-Net, which is a hybrid framework that integrates transformers-based global feature extraction with the nested U-Net architecture to better segment brain tumors in MRI scans. TransU2-Net uses transformers in the skip connections in order to retain spatial information. In addition to that, trans-formers-based multi-scale feature map fusion technique is employed for integrating high and low dimensional features. Furthermore, using self-attention mechanisms in the encoder and a deep U-Net decoder, TransU2-Net efficiently captures both global dependencies and fine-grained local details. The model was evaluated on the BraTS 2021 dataset, achieving Dice scores of 0.82 for enhancing tumor, 0.88 for whole tumor, and 0.86 for tumor core, thereby demonstrating its effectiveness in medical image segmentation [17].

The authors presented SwinVNETR, an architecture based on the Swin V-Net Transformer that includes a non-local block for the purpose of segmenting volumetric MRI brain tumors. Their approach integrates a 3D Swin Transformer to capture global relationships alongside a CNN encoder-decoder for extracting local features. Incorporating a non-local block strengthens the model's capacity to capture long-range dependencies, thereby improving segmentation accuracy. On the BraTS 2021 dataset, SwinVNETR achieved Dice scores of 0.84, 0.91, and 0.87 for enhancing tumor (ET), whole tumor (WT), and tumor core (TC), respectively. Additionally, the model makes use of Grad-CAM for better interpretability, which helps clinicians comprehend the segmentation outcomes [18].

Most recently, the authors proposed TDAConvAttention-Net, a brain tumor segmentation framework that explicitly integrates prior shape knowledge into a deep model. The method leverages Topological Data Analysis (TDA) with persistent homology to extract tumor shape features, which are concatenated with the raw input as an additional channel. To capture local and global context, the network employs a ConvNeXt-based U-Net backbone combined with multi-headed self-attention blocks in the skip connections. This hybrid design enables the model to effectively reduce topological errors, such as overlapping regions and broken connections, while preserving fine tumor morphology. Experiments on the BraTS2021 dataset demonstrated that incorporating shape priors significantly improved segmentation accuracy, outperforming state-of-the-art CNN- and transformer-based methods [19].

B. OPTIMIZATION APPROACHES

Jawad et al, introduced GSNet, a hybrid, end-to-end neural framework for multi-class segmentation aimed at automating brain tumor analysis. The architecture follows a five-level encoder–decoder design, incorporating attention at each stage, which enables it to capture rich 3D representations from MRI scans. Their evaluation was carried out on the BraTS 2019 and 2020 datasets [20].

In a related work, the authors developed a dynamic focal Dice loss to address challenges in multi-modal, multi-class glioma segmentation. This adaptive loss emphasizes smaller tumor subregions by optimizing the learning process for each class independently. Additionally, a symmetric attention branch was employed to localize tumors amidst irregular tissue patterns. Their approach, validated on the BraTS 2019 dataset, also utilized balanced patch sampling and extensive augmentations to improve generalization [21].

In a subsequent contribution, the authors enhanced a CNN-based segmentation model using an improved chimp optimization algorithm. Their pipeline normalized multi-modal MRI scans, extracted 17 discriminative features from segmented binary objects, and refined these features through optimization before training the CNN. This approach led to more reliable and precise tumor segmentation outcomes [22].

In another research, the authors have suggested an optimization-based machine learning method for classifying brain tumors that combines the metaheuristic feature selection with ensemble classifiers, minimizing feature redundancy while maximizing classification precision when analyzing MRIs [23]. Despite this study focusing on

ROI-based classification rather than voxel-wise tumor segmentation, it demonstrates the need to incorporate computational efficiency into brain tumor analysis workflows, which is a commonality between UFTrans and its parameterless Hadamard fusion with attention-driven feature suppression.

C. U-NET-BASED VARIANTS

Another study developed a framework for segmenting brain tumors based on the 2D U-Net architecture. This was achieved by using single 2D slices of volume MRIs, which decreased computational cost without compromising on accuracy. Preprocessing included normalization and slice selection, and the pipeline employed a lightweight encoder-decoder with skip connections. The approach achieved a Dice coefficient of 0.9358, accuracy of 0.9941, and sensitivity of 0.9897 on the BraTS 2020 dataset [24].

In a related study, the authors developed Edge U-Net, a deep convolutional neural network inspired by the U-Net architecture that emphasizes tumor boundaries. By incorporating boundary-aware information into both the decoder and the loss function, the model improved precision in delineating tumor regions. Tested on 3064 contrast-enhanced T1-weighted scans, it outperformed several competing methods [25].

In another contribution, the authors designed a residual 3D model for brain tumor segmentation, known as Deep Residual U-Net. Residual connections in the encoder phase alleviated vanishing gradient issues and allowed the network to leverage both high- and low-level details. Additional skip connections between convolutional layers and residual blocks further accelerated training. The model was evaluated on the BraTS 2021 dataset and demonstrated strong performance [26].

Finally, in a subsequent work, the authors proposed nnU-Net+, an enhanced version of the nnU-Net framework. By integrating residual blocks for better spatial feature extraction, attention gates to highlight tumor regions, and Hausdorff Distance loss for boundary refinement, the model achieved Dice scores of 0.79, 0.90, and 0.81 for enhancing tumor, whole tumor, and tumor core, respectively, on the BraTS dataset. These architectural refinements made nnU-Net+ a robust variant for medical image segmentation [27].

D. SUMMARY OF RELATED WORK STRENGTHS AND LIMITATIONS

The hybrid segmentation architectures summarized in Table 1 exhibit different trade-offs depending on their choices of local and global feature extractors, fusion strategies, and fusion stages. Methods such as UNETR and Swin-UNETR integrate transformer-based global features with CNN decoders mainly through skip connections at the decoder input, enabling global context incorporation but limiting early local–global interaction. Approaches including nnFormer and TransU2-Net perform more extensive feature fusion through skip attention or concatenation across multiple or skip stages, allowing richer feature integration at the expense of increased architectural complexity. In contrast, MVSI-Net restricts fusion to the bottleneck stage, reducing fusion overhead while limiting the influence of global information on fine-grained decoding. To address these limitations, the proposed UFTrans performs pre-decoder fusion by combining global features from a Vision Transformer with local features from a 3D attention U-Net using a Hadamard-based fusion strategy, enabling early and efficient interaction between complementary representations.

III. METHODOLOGY

This study presents a methodology for brain tumor segmentation that leverages a hybrid model combining transformer-based features with a 3D attention U-Net architecture. We detail the preprocessing steps, the model architecture, and the training procedures employed in our approach.

A. PREPROCESSING

1) MULTIMODALITY AND NORMALIZATION

MRI scans come with different types of sequences T1ce, T1, T2 and FLAIR as shown in Figure 1 and each sequence provides us with a different type of detail about tumors. A variety of modalities offer distinct sets of features. So, to get a comprehensive understanding of the subject we stacked Flair, T2, and T1ce. After stacking to ensure that the data has zero mean and unit variance, that helps model to stabilize the training process, each sample is standardized using z-score normalization:

$$Z_{i,j,k,d} = \frac{X_{i,j,k,d} - \mu_d}{\sigma_d} \tag{1}$$

where $Z_{i,j,k,d}$ is the normalized voxel value at position (i, j, k) in modality d and $X_{i,j,k,d}$ is the original voxel intensity at the same position. The mean intensity value μ_d for each modality is computed across all voxels in the 3D volume as:

$$\mu_d = \frac{1}{D \times H \times W} \sum_{i=1}^{D} \sum_{j=1}^{H} \sum_{k=1}^{W} X_{i,j,k,d} \tag{2}$$

Similarly, the standard deviation σ_d is calculated as:

$$\sigma_d = \sqrt{\frac{1}{D \times H \times W} \sum_{i=1}^{D} \sum_{j=1}^{H} \sum_{k=1}^{W} (X_{i,j,k,d} - \mu_d)^2} \tag{3}$$

where $D \times H \times W$ represents the total number of voxels in the 3D volume for each modality. Normalization ensures that voxel intensities are approximately scaled, improving the stability and convergence of the model during training.

2) TUMOR CENTER PATCHES

This approach mitigates the class imbalance in multiclass brain tumor segmentation. Brain tumors have sub-regions Edema, Necrosis, and Enhancing. It is observed in the BraTS dataset that these regions are highly imbalanced with respect to the background (no-tumor) region i.e. tumor voxels occupy

TABLE 1. Comparison of hybrid segmentation architectures.

Method	Local Extractor	Global Extractor	Fusion Strategy	Fusion Stage
UNETR [15]	CNN Decoder	Pure ViT	Skip connections	Decoder input
Swin-UNETR [12]	CNN Decoder	Swin Transformer	Skip connections	Decoder input
nnFormer [13]	ConvNet	Local-Global Transformer	Skip attention	Multiple stages
TransU2-Net [17]	U2-Net	Transformer	Concatenation	Skip connections
MVSI-Net [16]	U-Net	Multi-view Attention	MSI module	Bottleneck only
UFTrans (Ours)	**3D Attention U-Net**	**Vision Transformer**	**Hadamard**	**Pre-decoder**

only a small fraction of the brain volume. Such imbalance can lead to poor segmentation performance and unnecessary computation when training on predominantly background voxels.

We adopt and re-implement the methodology from Tumor Center Patching for enhanced medical image segmentation paper [28] to generate a tumor-focused $128 \times 128 \times 128 \times 3$ input volume from the original $240 \times 240 \times 155 \times 3$ multi-channel MRI Figure 2. In this pipeline, tumor localization is performed on the FLAIR channel only, a superlevel-set filtration over FLAIR intensities is constructed and persistent homology is computed for H_0 (connected components). Components with high persistence are retained to suppress threshold-sensitive noise and produce a tumor-candidate response/mask then we apply 3D connected component analysis (CCA) to the retained candidate mask, remove small spurious regions, and select the most tumor-like component according to the criteria described by author. The centroid of the selected component is used as the patch center, and a fixed 128^3 region is cropped from the original 3-channel MRI volume to form the final $128 \times 128 \times 128 \times 3$ patch. Since this preprocessing module is adopted from prior work, we refer readers to Tumor Center Patching for enhanced medical image segmentation [28] for full algorithmic and parameter details.

During inference, ground truth labels are unavailable; therefore, patch extraction must rely solely on the input MRI intensities. We use a two-stage strategy. First, we estimate the tumor centroid from the raw FLAIR channel using the same TDA+H_0 persistent connected components and 3D CCA procedure(no segmentation masks are required). A fixed 128^3 patch is then cropped from the original 3-channel MRI volume centered at the estimated centroid and fed to UFTrans.

To ensure robustness in cases where the localization stage fails (e.g., extremely low contrast or fragmented responses), we employ a sliding-window fallback over the full $240 \times 240 \times 155$ volume using 128^3 windows with 50% overlap. UFTrans predictions from overlapping windows are aggregated by voxel-wise averaging of softmax probabilities, followed by an arg max operation to obtain the final segmentation.

3) DATA AUGMENTATION

Data augmentation was applied on the training set to enhance the training data's diversity and also to achieve improved generalization in the model. This step is particularly important in medical image segmentation tasks to account for variations in anatomical structures, imaging conditions, and scanner differences. The augmentation pipeline was designed to include a variety of spatial transformations, for the simulation of realistic variations without compromising the integrity of the medical data.

To enhance the model's invariance with regard to orientation changes and improve validity against anatomical variations, many data augmentation techniques were applied to the 3D volumes. These incorporate random rotations to address discrepancies in orientation, elastic transformations to simulate realistic deformations of brain tissue, and random flips along spatial axes to introduce symmetric variations that facilitate the model's ability to segment tumors from diverse perspectives. Additionally, random scaling operations simulated variations in tumor size, enhancing the model's ability to handle tumors of different dimensions. In contrast, random 90-degree rotations were incorporated to further increase spatial variability within the dataset. This augmentation process ensured that the training data adequately covered potential variations in clinical scenarios, helping the model to generalize well to unseen data.

B. MODEL ARCHITECTURE

1) TRANSFORMER FEATURES EXTRACTION

To extract long-range dependencies and contextual information, we implemented a vision transformer [29] for 3D MRI data. This model directly processed 3D medical images by breaking them into smaller patches, converting them to embedding, and finally leveraging the transformer encoder to extract spatial dependencies across the entire image using these embeddings. Transformers have also shown remarkable generalization across different domains, highlighting their adaptability and making them useful for medical imaging tasks [30]. However, training large-scale transformer models often presents substantial GPU memory challenges, which requires adopting more efficient variant of this architecture [31].We trained from scratch without ImageNet pre-training, as domain-specific 3D medical imaging features differ significantly from 2D natural images.

Firstly, to reduce the input's complexity and make the data manageable for the transformer encoder, the input MRI

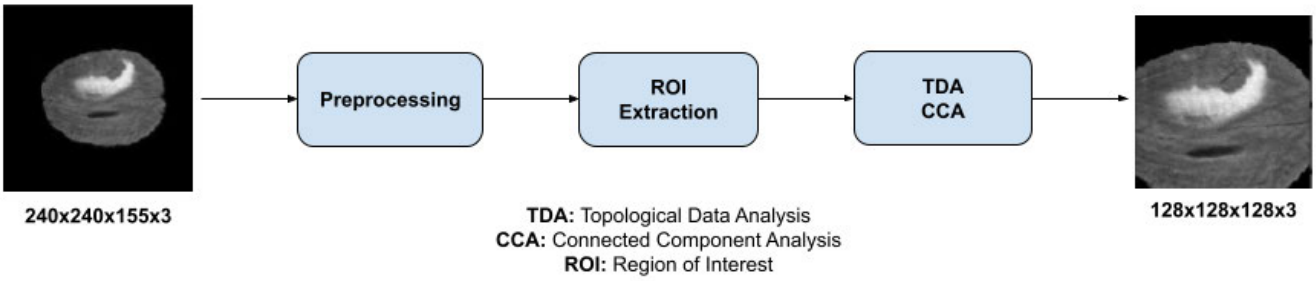


FIGURE 2. Overall workflow for tumor center patching. The input MRI volume (240×240×155×3) undergoes preprocessing, ROI extraction, and feature enhancement using Topological Data Analysis (TDA) and Connected Component Analysis (CCA). The final output is a refined MRI volume of size 128×128×128×3, highlighting the tumor region for further analysis.

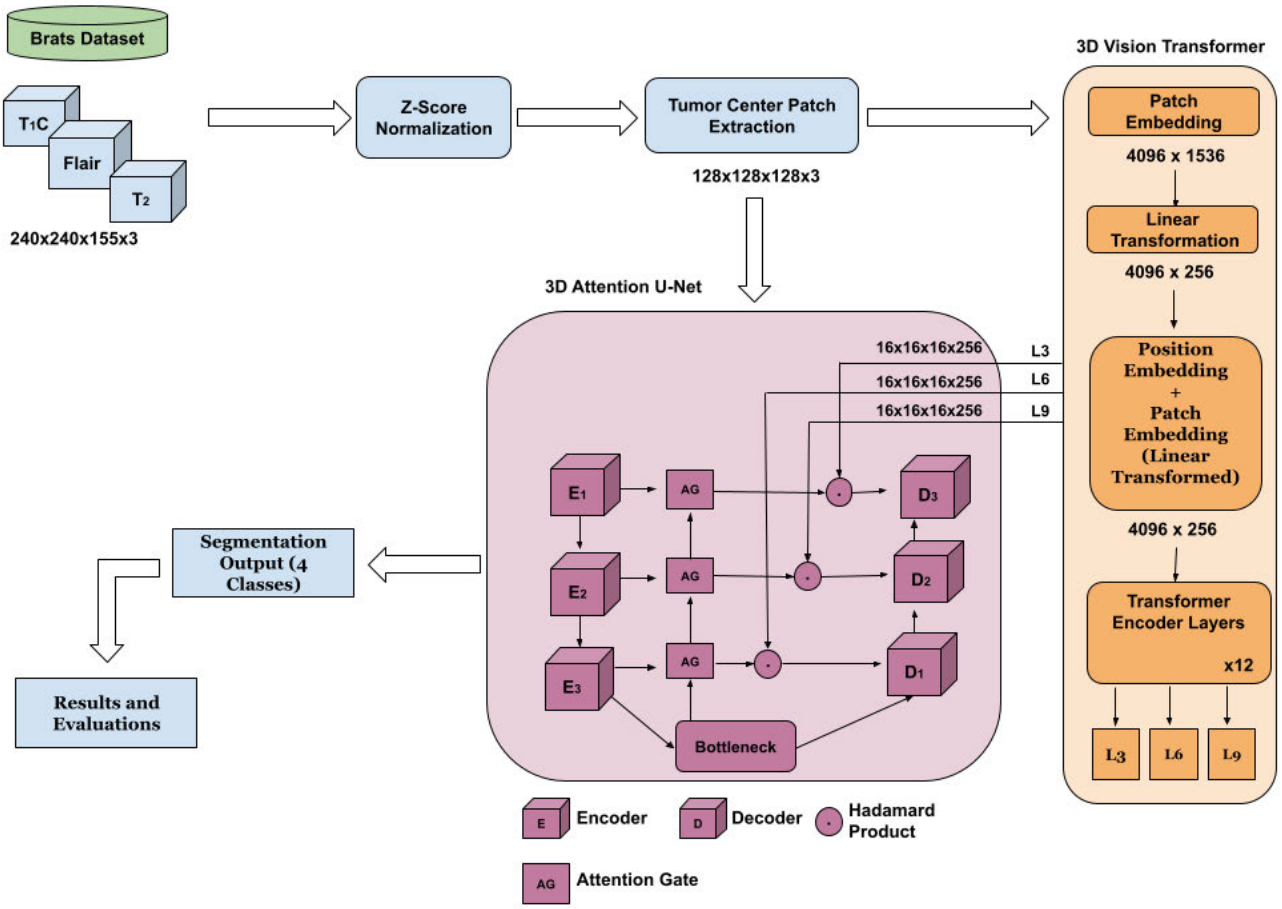


FIGURE 3. Overview of the proposed 3D attention U-Net architecture. The encoder uses convolutional layers to extract local features, while the transformer module captures global dependencies. Skip connections are enriched with attention weights to improve feature refinement. Feature fusion is performed using the Hadamard product to combine local and global features.

volume consists of 3 channels (corresponding to different modalities) split into non-overlapping patches of dimension 8 × 8 × 8. These patches are further embedded into a lower-dimensional feature space through a linear transformation, that prepares the data for subsequent processing by the transformer. Furthermore, as transformer models do not inherently encode spatial information, positional embeddings are added to the patch embeddings to ensure that the model retains information about each patch's location in the original 3D volume.

Our data is ready to be fed into a transformer as embeddings. In our case, these embeddings are fed into a series of 12 transformer encoder layers. To capture long-range dependencies, each encoder layer applies multi-head self-attention followed by a multilayer perceptron having 512-dimensional MLP hidden layer and 0.2 dropouts,

enabling the model to attend to different regions of the image simultaneously. After processing through the transformer layers, the model leverages the learned global representations for further analysis.

At last, the output from the transformer encoder's different layers is reshaped using transposed convolution and returned to fused with 3D attention U-Net.

2) PATCH EXTRACTION

The 3D convolution layer divides the input volume of shape $128 \times 128 \times 128 \times 3$ into non-overlapping 3D patches. The convolutional operation applies a kernel and stride with a size same as patch size P. For this model, a patch size of $P = 8$ is used, indicating that each patch corresponds to an $8 \times 8 \times 8$ volume. Additionally, a stride of $S = 8$ ensures that the patches do not overlap.

We set the 3D patch size to P=8 (yielding $16 \times 16 \times 16 = 4096$ tokens) to balance local anatomical detail and computational efficiency: reducing to $P = 4$ increases the token count by $8\times$ ($32^3 = 32768$ tokens), while increasing to $P = 16$ reduces tokens to $8^3 = 512$ but risks losing fine-grained tumor boundary cues.

The convolutional operation extracts 3D patches from the input volume, where each patch has a flattened dimension of $P \times P \times P \times C$. Mathematically, we can calculate the number of extracted patches as follows:

$$\begin{aligned}\text{Number of Patches} &= \left(\frac{D}{P}\right)\left(\frac{H}{P}\right)\left(\frac{W}{P}\right) \\ &= \left(\frac{128}{8}\right)\left(\frac{128}{8}\right)\left(\frac{128}{8}\right) = 16^3 = 4096\end{aligned} \tag{4}$$

Therefore, after patch extraction, the output reshapes to a tensor with dimensions (batch_size, num_patches, $P \times P \times P \times C$).

In this model, each patch's flattened vector length is $8 \times 8 \times 8 \times 3 = 1536$.

After patch extraction, the output tensor is reshaped to (batch_size, 4096, 1536) (for a single volume, it becomes (4096, 1536)), where each of the 4096 patches has 1536 values.

3) LINEAR TRANSFORMATION AND POSITIONAL EMBEDDINGS

The following represents the linear transformation of the patches:

$$\text{Embeddings} = \text{Linear}(x) = x \times W + b \tag{5}$$

where $x \in \mathbb{R}^{\text{num_patches} \times 1536}$ represents the flattened patches (for one volume), $W \in \mathbb{R}^{1536 \times \text{hidden_dim}}$ is the learnable weight matrix, and $b \in \mathbb{R}^{\text{hidden_dim}}$ is the bias term.

After applying a linear transformation on patches, each vector of size 1536 is mapped to a new vector of size equal to the hidden dimension (256). As transformer models are not inherently aware of the spatial positions of patches, positional embeddings are added:

$$x_{\text{final}} = \text{Patch Embeddings} + \text{Positional Embeddings} \tag{6}$$

We employ learned 3D positional encodings with a trainable parameter of shape (4096, 256), enabling the model to retain spatial information about each patch's location in the original 3D volume. Once the positional embeddings are added, the output tensor is ready to input into the transformer encoder layers. This aims to capture long-range dependencies across these patches, allowing the model to understand contextual information within the 3D volume. The transformer encoder consists of multi-head self-attention followed by a feed-forward network.

4) MULTI-HEAD SELF-ATTENTION

Each layer of the transformer encoder utilizes multi-head self-attention, allowing the model to capture information from multiple regions of the input sequence simultaneously. The attention score is calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{7}$$

In the multi-head mechanism, Q, K, and V matrices are learned for each head (8 heads), enabling the network to effectively capture varied relationships within the data. We employ 12 transformer encoder layers with MLP hidden dimension of 512.

5) FEED FORWARD NEURAL NETWORK

After the self-attention step, each transformer encoder layer has a feed-forward network (FFN) with 2 fully connected layers and GELU activation function. Each sub-layer is wrapped with residual connection and layer normalization. The residual connections help maintain the original input information, ensuring stable learning.

Lastly, the output from the transformer encoder is reshaped from (B, 4096, 256) to (B, 256, 16, 16, 16) using transposed convolution and returned to be fused with 3D attention U-Net.

6) 3D ATTENTION U-NET

The 3D attention based U-Net model extracts localized spatial features. This modified version of the standard U-Net [32] can directly process 3D MRI volumes. Attention gates [33] are incorporated in 3D U-Net to refine the features passed using skip connections. The attention module selectively concentrates on the most relevant features while disregarding irrelevant features.

7) ENCODER

The encoder block in 3D attention U-Net extracts feature representations from the input volume at multiple spatial scales. It is composed of two major components a convolutional block and max pooling layer. The convolutional block contains two 3D convolution layers using $3 \times 3 \times 3$ kernel and a padding value of 1, succeeded by a ReLU and batch normalization function. The convolutional

layers capture spatial features from the input data, while batch normalization stabilizes the learning process, and the ReLU activation introduces non-linearity. Post-convolutional operations, a max pooling layer with $2 \times 2 \times 2$ kernel size is applied to downsample the information maps by half in each dimension. This operation reduces the spatial dimensions, enabling the model to focus on abstract and higher-level features as its depth increases.

8) ATTENTION GATE

The attention gate allows the model to concentrate on essential regions of the feature map by applying a learned weighting to the skip connection from the encoder. This mechanism can greatly enhance segmentation score, particularly in cases where the objects of interest take up a small part of the input volume.

a: WEIGHTING OF INPUTS

Input of the attention gate is derived from output of the bottleneck and the skip connections. It applies separate convolutions to both inputs, reducing them to a common number of output channels.In order to promote stable training, batch normalization is applied after each convolution.

b: ELEMENT-WISE ADDITION AND ACTIVATION

Element-wise addition of the convolution outputs is performed, followed by ReLU function. This allows the model to combine the features in a non-linear manner.

c: SIGMOID OUTPUT

The final output of the attention gate is obtained by passing the result through a $1 \times 1 \times 1$ convolution layer followed by sigmoid activation, resulting in a value between 0 and 1 for each spatial location. Element-wise multiplication of these values with the skip connection output modulates the feature map based on the learned attention weights. Figure 4 provides the workflow of the attention mechanism. Attention gates are applied to the encoder skip features at scales $E_1(128^3)$, $E_2(64^3)$, and $E_3(32^3)$ *before* Hadamard fusion with the corresponding transformer maps (F_{128}, F_{64}, F_{32}), so that only salient skip responses are forwarded to the decoder.
Following the additive attention mechanism shown in Fig. 4, given a skip feature map x and a gating feature map g from a deeper layer, the AG computes an intermediate attention score:

$$q = \mathrm{ReLU}(\mathrm{Conv}_{1\times1\times1}(x) + \mathrm{Conv}_{1\times1\times1}(g) + b)\,, \tag{8}$$

which is converted into an attention mask:

$$\alpha = \mathrm{Resample}\big(\sigma\big(\mathrm{Conv}_{1\times1\times1}(q)\big)\big)\,, \tag{9}$$

and finally reweights the skip feature via element-wise multiplication:

$$x' = \alpha \odot x. \tag{10}$$

In Eqs. (8)–(10), $\mathrm{Conv}_{1\times1\times1}(\cdot)$ denotes a learnable $1 \times 1 \times 1$ convolution for channel alignment, b is a bias term, $\sigma(\cdot)$ is the sigmoid function, $\odot$ denotes the Hadamard (element-wise) product, and Resample$(\cdot)$ adjusts α to match the spatial resolution of x when needed. Here, x corresponds to a skip feature (E_1, E_2, E_3), and g denotes the bottleneck (or deeper decoder) feature resampled to the same spatial size as x.

9) DECODER

Using an upsampling operation, the decoder block is responsible for recovering spatial resolution in the combined feature maps. The decoder block begins with an upsampling operation using trilinear interpolation. The scale factor for upsampling is 2, doubling the feature maps' spatial resolution. Features from bottleneck are upsampled and concatenated with attention-weighted features.

10) FEATURE FUSION

Our proposed model integrates the features of the transformer and the 3D attention U-Net between the skip connections. Transformer features extracted at layers L_3, L_6, and L_9 are reshaped to $F_t^{(k)} \in \mathbb{R}^{B\times256\times16\times16\times16}$ and progressively upsampled using transposed convolution to $\{F_{32}, F_{64}, F_{128}\}$, these multi-scale maps are Hadamard-fused with the corresponding 3D Attention U-Net encoder features $\{E_3(32^3), E_2(64^3), E_1(128^3)\}$ at each skip level. These fused features are utilized in the decoder phase to produce the segmented output. Transformer captures long-range dependencies, and its features are merged with those from the 3D Attention U-Net via skip connections. The fusion process employs the Hadamard product, which helps emphasize highly relevant feature interactions. This hybrid representation effectively fused global contextual information from the transformer with local spatial features from the U-Net, improving tumor segmentation score. Figure 3 illustrates the overall architecture of our proposed model.

Given transformer features F_T and attention-gated U-Net features F_U, the Hadamard product $F_T \odot F_U$ acts as a parameter-free multiplicative agreement gate: spatial locations where both branches assign high activations are amplified, while locations where either branch produces low activation are driven toward zero. This produces a selective intersection of evidential support from both branches.

On the other hand, due to concatenation $[F_T, F_U]$ the channel dimension gets doubled ($C \to 2C$), without doing any explicit feature interaction but increases the number of downstream parameters, whereas addition $F_T + F_U$ simply adds activations together and thus allows strong single-branch noise to propagate unchecked. The multiplication gate concept is in line with the mechanism used in Squeeze-and-Excitation Networks [34], by reassigning weights to channels according to their importance scores and self-attention by using query-key dot product as a multiplicative gate.The Hadamard fusion ablation results show that (Table 8) our approach is better than both concatenation and addition by +2% mean Dice consistently across all three subregions.

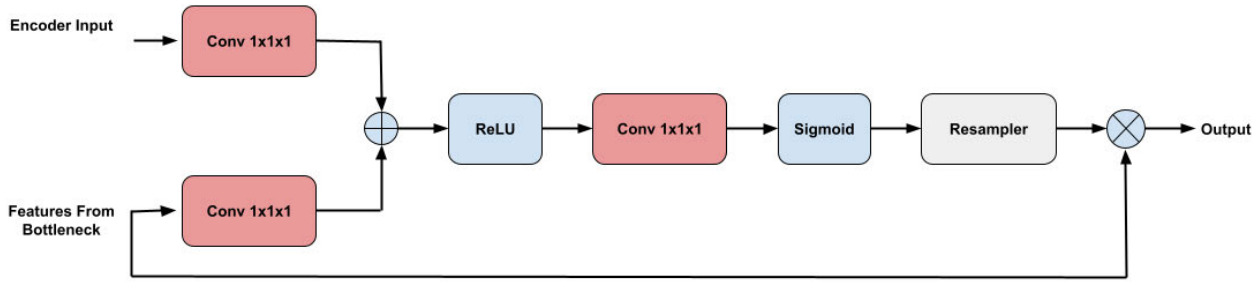


FIGURE 4. Workflow of the attention mechanism in 3D attention U-Net: The attention gate selectively refines features byweighting inputs from the encoder and bottleneck. The mechanism suppresses irrelevant information and enhances relevantspatial regions, aiding in precise tumor segmentation.

TABLE 2. UFTrans.

Input: MRI volume $V \in \mathbb{R}^{240\times240\times155\times3}$
Output: Segmentation mask $M \in \{0,1,2,4\}^{128\times128\times128}$
Preprocessing:
1. Apply z-score normalization per modality.
2. Extract a tumor-centered patch $P \in \mathbb{R}^{128\times128\times128\times3}$ via TDA+CCA.
Forward Pass:
3. **Transformer Branch (3D ViT):**
3a. Partition P into 4096 non-overlapping 3D patches of size 8^3.
3b. Flatten each patch to a 1536-d token $(8^3 \times 3)$, forming $X \in \mathbb{R}^{B\times4096\times1536}$.
3c. Linearly project tokens to 256-d embeddings: $Z \in \mathbb{R}^{B\times4096\times256}$.
3d. Add learned 3D positional embeddings and pass through 12 transformer encoder layers.
3e. Extract intermediate features at layers L_3, L_6, and L_9.
3f. Reshape tokens to a 3D volume: $F_t^{(k)} \in \mathbb{R}^{B\times256\times16\times16\times16}$, $k \in \{3,6,9\}$.
3g. Build multi-scale transformer maps (used for skip fusion) via stride-2 transposed conv.:
$F_{16} \leftarrow F_t^{(9)} \in \mathbb{R}^{B\times256\times16\times16\times16}$ *(base map for upsampling only)*
$F_{32} = \text{Deconv}_{\times2}(F_{16}) \in \mathbb{R}^{B\times256\times32\times32\times32}$
$F_{64} = \text{Deconv}_{\times2}(F_{32}) \in \mathbb{R}^{B\times256\times64\times64\times64}$
$F_{128} = \text{Deconv}_{\times2}(F_{64}) \in \mathbb{R}^{B\times256\times128\times128\times128}$.
4. **U-Net Branch (3D Attention U-Net):**
4a. Encode P with 3 downsampling stages to obtain skip features $E_1(128^3)$, $E_2(64^3)$, $E_3(32^3)$ and bottleneck $B_n(16^3)$.
4b. Apply attention gates to skip features *before fusion* to obtain E_1', E_2', E_3'.
5. **Multi-scale Hadamard Fusion (Fusion Points):**
Fuse at three skip scales (as in Fig.3):
$\tilde{E}_3 = E_3' \odot \phi(F_{32}), \quad \tilde{E}_2 = E_2' \odot \phi(F_{64}), \quad \tilde{E}_1 = E_1' \odot \phi(F_{128})$
where $\phi(\cdot)$ denotes channel alignment via $1\times1\times1$ conv if required.
6. **Decoder:** Upsample and decode using fused skips $\{\tilde{E}_3, \tilde{E}_2, \tilde{E}_1\}$.
7. **Output:** Apply softmax for 4-class voxel-wise prediction.
Training: Optimize $\mathcal{L} = 0.5\,\mathcal{L}_{Dice} + 0.5\,\mathcal{L}_{Focal}$.

C. OVERALL ALGORITHM

Table 2 represents the overall training and inference procedure for UFTrans.

D. LOSS FUNCTION

1) DICE LOSS

The primary loss function used in medical image segmentation is dice loss [35]. This function measures the resemblance between two sets. This function is most commonly used in segmentation tasks to measure the resemblance be- tween two sets. The dice coefficient is defined as twice the intersection of two sets, one actual and the other predicted, divided by the sum of these two sets, and we get our loss by subtracting 1 from the dice coefficient because dice coef- ficient range from 0 to 1. equation 11 shows the formula for dice loss.

$$\text{Dice Loss}_{\text{multiclass}} = 1 - \frac{1}{C}\sum_{c=1}^{C}\frac{2\sum_{i=1}^{N} p_{i,c}g_{i,c} + \epsilon}{\sum_{i=1}^{N} p_{i,c} + \sum_{i=1}^{N} g_{i,c} + \epsilon} \tag{11}$$

where $c \in \{1,\ldots,C\}$ indexes the segmentation classes and $i \in \{1,\ldots,N\}$ indexes voxels in the training sample (i.e., the 128^3 patch; in practice, the sums can be aggregated over the batch). $p_{i,c} \in [0,1]$ denotes the softmax probability that voxel i belongs to class c, and $g_{i,c} \in \{0,1\}$ is the corresponding one-hot ground-truth indicator. ϵ is a small constant for numerical stability.

2) FOCAL LOSS

Focal loss [36] is used to address the class imbalance problem. In our case, the data is highly imbalanced, with some tumor regions covering more area than others. Equation 12 shows how we can calculate the focal loss. Using a modification of the standard cross-entropy loss, the focal loss is obtained.

$$\mathcal{L}_{Focal} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}\alpha_c \cdot (1-p_{i,c})^{\gamma} \cdot g_{i,c} \cdot \log(p_{i,c}) \tag{12}$$

where N is the number of voxels, C is the number of classes, $p_{i,c} \in [0,1]$ denotes the predicted probability (after softmax) of voxel i belonging to class c, and $g_{i,c} \in \{0,1\}$ is the corresponding one-hot ground-truth indicator. α_c is the weighting factor for class c (set to handle class imbalance) and $\gamma = 2$ is the focusing parameter that adjusts the importance given to hard-to-classify examples.

3) COMBINED LOSS FUNCTION

Combining Focal Loss and Dice Loss provides complementary benefits for medical image segmentation tasks. Focal Loss proves helpful in the case of class imbalance by focusing on hard-to-classify regions, suppressing well-classified background pixels, and emphasizing on complex foreground

regions, such as tumors. This will enable the network to learn from the challenging samples. Dice Loss is one of the best approaches that can be used to segment small objects since it focuses on overlap and precision of the object's boundaries. Moreover, it directly evaluates the similarity between the model's pre- dictions and the annotated ground truth. These losses complement each other in the sense that they combine global classification (by Focal Loss) and local refinement (by Dice Loss) to get accurate segmentation, especially when it comes to imbalanced datasets where small foreground regions need more attention without being overpowered by the background. The equation for their combined weight is given in 13 below:

$$\mathcal{L}_{total} = \lambda_1 \cdot \mathcal{L}_{Dice} + \lambda_2 \cdot \mathcal{L}_{Focal} \tag{13}$$

with $\lambda_1 = 0.5$ and $\lambda_2 = 0.5$ as fixed weights in all experiments. The focusing parameter $\gamma = 2.0$ and class weights α_c are calculated automatically based on the inverse frequency of classes. Smoothing factor parameter $\epsilon = 10^{-7}$ is used in the Dice loss function to avoid division by zero. This combination ensures that both Focal and Dice losses complement each other.

IV. EXPERIMENTS AND RESULTS

The following section describes the dataset used in the experiments for training and testing, evaluation metrics used to measure the performance of the proposed technique, the implementation details, and comparisons with state-of-the-art methods.

A. DATASET

The proposed model is trained and validated on BraTS 2021 dataset [37], [38], [39]. This dataset contains MRI data of 1251 glioma patients each sample is a 3D volume with dimensions 240 × 240 × 155. The BraTS 2021 dataset comprises multimodal MRI sequences saved as nifty files (.nii.gz), representing different aspects of brain tumor imaging. These include T2-weighted (T2), T2, T1, post-contrast T1-weighted (T1Gd) and Fluid Attenuated Inversion Recovery (T2-FLAIR) MRIs. The above mentioned scans were acquired using various clinical procedures and scanning equipment from 19 different institutions. Each MRI scan in dataset has been meticulously segmented manually by 1-4 raters, under a unified protocol, with the final labels validated by senior neuroradiologists. The segmentation task defines three regions of interest: peritumoral edema (label 2), gadolinium-enhancing tumor (label 4), and the necrotic or non-enhancing tumor core (label 1). This labeling strategy has been consistently applied across different BraTS challenges. Prior to release, all scans are preprocessed through co-registration to a common anatomical space, resampled to an isotropic resolution of 1 mm^3, and subjected to skull stripping. These standardized procedures ensure that the data are ready for analysis and facilitate comparability across different scans and research studies. Different imaging modalities capture distinct types of significant information about the tumor. The FLAIR modality highlights the edema (ED) region, providing clear boundaries between the normal and tumor regions. Meanwhile, the T1ce modality emphasizes the tumor core (TC) with high contrast. We split the BraTS 2021 training cases at the *patient level* into 80% for training and 20% as a held-out validation set for model selection and reporting, ensuring no subject appears in more than one subset (no data leakage across splits). Since BraTS test labels are not publicly available, standardized benchmarking on the official BraTS validation/test server is left as future work.

TABLE 3. Dataset details used in experimentation.

Attribute	Description
Dataset Name	BraTS 2021
Total Patients	1251 glioma patients
Volume Dimensions	(240, 240, 155)
File Format	NIfTI (.nii.gz)
MRI Modalities	T1, T1Gd, T2, T2-FLAIR
Segmentation Labels	1: Necrotic, 2: Edema, 4: Enhancing Tumor

TABLE 4. Model configuration and parameters.

Configuration Name	Values
Training Data	80%
Held-out Validation Data	20%
Input Modalities (T1ce, T2, Flair)	3
Batch Size	2
Number of Epochs	100
Optimizer	Adam
Learning Rate	0.0001
Dropout	0.2
Number of Encoders	3
Number of Decoders	3
Patch Size in Transformer	8×8×8
Embedding Vector Size	256
MLP Dimension Transformer	512
Number of Layers in Transformer	12
Number of Attention Heads	8
Loss Function	Dice loss + focal loss
Output Layer Activation	Softmax

B. IMPLEMENTATION DETAILS

Our deep learning model and other baseline approaches were trained and evaluated under identical environment and settings on an Intel i9 13-gen processor with 128 GB of memory and NVIDIA RTX A6000 48 GB GPU. The model converged after approximately 80 epochs, with early stopping applied (patience=15 based on validation Dice). Training time was approximately 9.2 hours for the full 100 epochs. Inference time is approximately 71ms per volume on the same hardware. GPU memory usage is approximately 5GB during inference and 24GB during training with batch size 2. A detailed summary of the hyperparameters and training settings is provided in Table 4.

C. EVALUATION METRICS

1) DICE SIMILARITY COEFFICIENT

The dice similarity coefficient is used to compare and evaluate the performance of our model because it is widely used in segmentation evaluation throughout the literature,

specifically in medical image segmentation. equation 14 shows the formula to calculate the dice similarity coefficient.

$$\mathrm{DSC}_{\mathrm{multiclass}} = \frac{1}{C}\sum_{c=1}^{C}\frac{2\sum_{i=1}^{N} p_{i,c}g_{i,c}}{\sum_{i=1}^{N} p_{i,c} + \sum_{i=1}^{N} g_{i,c}} \tag{14}$$

Furthermore, to compare results with other models, we examined three types of regions: the tumor core, a combination of enhancing and necrosis regions, and the whole tumor, providing an overall score for all the regions present in the volume (Necrosis, Enhancing, Edema). The last type is the enhancing region.

D. RESULTS AND DISCUSSION

In this section, we conduct a detailed analysis of the quantitative metrics and qualitative insights from our model, comparing them to leading state-of-the-art models to highlight our approach's strengths and weaknesses.

1) QUANTITATIVE RESULTS

We evaluated UFTrans model on the BraTS 2021 dataset, obtaining cutting-edge segmentation accuracy. The model reached Dice scores of **0.90** for the whole tumor (WT), **0.91** for enhancing tumor (ET), and **0.89** for tumor core (TC) on the validation set. These outcomes surpass those of several recent segmentation approaches that utilize deep learning methods. A higher Dice score reflects better alignment between the ground truth mask and the predicted tumor segmentation. Table 5 presents the quantitative comparisons of our approach with other SOTA approaches. In addition to the Dice score, we also use the 95th percentile Hausdorff distance (HD95) as a measure of evaluation for boundary accuracy, with lower values indicating better boundary precision. The following table 6 gives HD95 results for UFTrans on WT, ET, and TC. Moreover, we present here the comparison of computational complexity to contextualize performance with respect to model size and runtime.

Table 7 reports parameter count, training time, inference time, and GPU memory usage for UFTrans and representative baselines.

To analyze the contribution of fusion strategy and key architectural components, we report ablation results in Table 8 and Table 9. These results show that the Hadamard fusion and attention-based components contribute to improved segmentation performance across WT, ET, and TC. We performed further investigation on the interaction effects between the components to ensure they operate synergistically. As shown in table 10, there is a clear interaction effect of the vision transformer depth and Hadamard fusion stages, in which the multi-scale fusion requires deep global context (i.e., 12-layer vision transformers) to efficiently guide the decoding process, while shallow transformers are not capable of exploiting the potential of multi-stage fusion. Furthermore, the increase in the number of depth layers up to 18 results in diminishing returns, implying that 12 depth layers offer the best feature abstraction. Moreover, Table11 presents the interplay between our tumor-centered patching strategy and the 3D Attention Gates. It is evident that even when tumor-centered patching successfully isolates the region of interest, spatial attention gates remain crucial in filtering localized noise within those targeted patches. This confirms that these modules are not redundant, but rather act as complementary mechanisms for refining local and global representations.

2) COMPARISON WITH CNN-BASED MODELS

Convolutional architectures such as nnU-Net and MedNeXt achieved remarkable accuracy in medical image segmentation by taking advantage of convolutional feature extractors for localized spatial learning. These networks, however, struggle to capture long-range dependencies, which are critical in segmenting large and heterogeneous tumors. nnU-Net, a widely used self-configuring CNN model, adapts its architecture to different datasets and achieves competitive results. Even though this model performs efficiently, it fails to incorporate contextual information within the tumor boundaries due to local convolution operations only. To overcome this problem, we introduced a transformer encoder component in our design to improve tumor localization by adding global context without sacrificing spatial resolutions in a 3D attention U-Net. This leads to improved boundary precision, particularly in cases where tumors exhibit diffuse or infiltrative growth patterns, as reflected by the HD95 values in Table 6. Similarly, MedNeXt is a modern CNN architecture built upon the principles of ConvNeXt and includes residual convolutional layers and large kernel convolutions to facilitate multi-scale representation learning. Although this network outperforms other conventional CNN models by providing better boundary accuracy, it lacks mechanisms for incorporating global contextual features. To solve this issue, the hybrid CNN- transformer model architecture, utilizing Hadamard fusion and ensuring an effective balance between CNN and transformer feature contributions (Table 8). This enables optimized segmentation score across all tumor subregions, leading to more precise and reliable tumor delineation.

3) COMPARISON WITH TRANSFORMER-BASED MODELS

Transformer-based segmentation models, such as UNETR, Swin UNETR, nnFormer, and MedNeXt, have played an important role in the development of feature extraction in medical imaging. However, their dependence on self-attention operations can result in a lack of local feature conservation, thus negatively impacting their segmentation accuracy around the boundaries of tumors. UNETR, introduced by [15], employs a Vision Transformer (ViT) encoder to capture long-range dependencies within MRI volumes, enhancing tumor localization but struggling with fine-grained boundary segmentation, achieving a Dice score of 80% for whole tumor segmentation on the BraTS 2021 dataset (Table 5). Swin UNETR [12], an extension of UNETR, introduces hierarchical self-attention using Swin Transformer

TABLE 5. Comparative analysis on BraTS 2021 (dice score).

Method	Whole Tumor (WT)	Enhancing Tumor (ET)	Tumor Core (TC)
MVSI-Net [16]	0.85	0.86	0.78
TransU2-Net [17]	0.88	0.82	0.86
nnU-Net+ [27]	**0.90**	0.79	0.81
UNETR [15]	0.80	0.82	0.77
SwinUNETR [12]	0.88	0.88	0.87
nnFormer [13]	0.85	0.85	0.83
MedNeXt [14]	0.88	0.88	0.86
UFTrans (Proposed)	**0.90**	**0.91**	**0.89**

TABLE 6. UFTrans hausdorff distance 95% (HD95).

Region	HD95 (mm)
Whole Tumor (WT)	2.96
Enhancing Tumor (ET)	2.23
Tumor Core (TC)	2.87

TABLE 7. Computational complexity analysis (memory is reported as train/infer GB).

Method	Params (M)	Train (hrs)	Infer (ms)	Mem
nnU-Net [27]	31.2	18	3200	12
Swin-UNETR [12]	62.1	36	5800	24
UFTrans	**11**	**9.2**	**71**	**24/5**

TABLE 8. Ablation study on fusion strategies (dice score).

Fusion Mode	WT	ET	TC	Mean
No Fusion	0.87	0.88	0.86	0.87
Concatenation	0.88	0.89	0.87	0.88
Addition	0.88	0.90	0.87	0.88
Hadamard (Proposed)	**0.90**	**0.91**	**0.89**	**0.90**

TABLE 9. Ablation study on architecture components.

Configuration	WT	ET	TC
3D U-Net (no attention)	0.86	0.87	0.85
3D Attention U-Net	0.87	0.88	0.86
UFTrans (no attention gates)	0.88	0.89	0.87
UFTrans Full	**0.90**	**0.91**	**0.89**

encoders for multi-scale feature extraction, achieving Dice scores of 88% (WT), 87% (TC), and 88% (ET) on BraTS 2021 (Table 5). Despite its high accuracy, Swin UNETR's complex hierarchical design increases computational overhead (Table 7). nnFormer, developed by [13], improves volumetric segmentation using 3D self-attention and skip attention mechanisms to facilitate feature interaction between encoder and decoder layers. Still, it lacks the ability to preserve fine local details, often resulting in boundary misalignment. UFTrans addresses these limitations by combining a 3D attention U-Net that refines local spatial feature extraction, ensuring superior segmentation precision. Additionally, our approach employs Hadamard-based feature fusion to enhance the mutual representation of local and global features, leading to improved segmentation accuracy, sharper tumor boundaries (Table 6), and faster convergence.

4) COMPARISON WITH HYBRID MODELS

Hybrid models such as TransU2-Net and MVSI-Net have been developed to integrate convolutional and transformer-based architectures for improved medical image segmentation. TransU2-Net enhances global feature representation by incorporating transformer blocks within the skip connections of a U-Net framework. However, the dependence of the transformer block output on the U-Net encoder input can impact segmentation performance, as any inaccuracies in early feature extraction may propagate through subsequent layers. Additionally, the use of transformer blocks in every skip connection influences feature refinement, leading to potential inconsistencies in learned representations across different scales. MVSI-Net employs multi-view attention alongside multi-scale feature interaction to capture hierarchical contextual information, thereby enhancing segmentation performance. However, the sequential processing of local and global features in MVSI-Net rather than their integrated fusion may limit the network's ability to fully utilize both types of information simultaneously. UFTrans addresses these challenges by leveraging a Hadamard-based fusion strategy (Table 8), ensuring effective integration of local and global representations at multiple levels within the network. This approach enhances segmentation score by refining the interaction between convolutional and transformer-based feature extraction, leading to more precise tumor delineation.

TABLE 10. Interaction of transformer depth and fusion stage (dice score).

Transformer Depth	Bottleneck Only	Multi-Scale Fusion
6 Layers	0.83	0.85
12 Layers	0.86	**0.90**
18 Layers	0.88	0.89

TABLE 11. Interaction of attention gates and patching strategy (dice score).

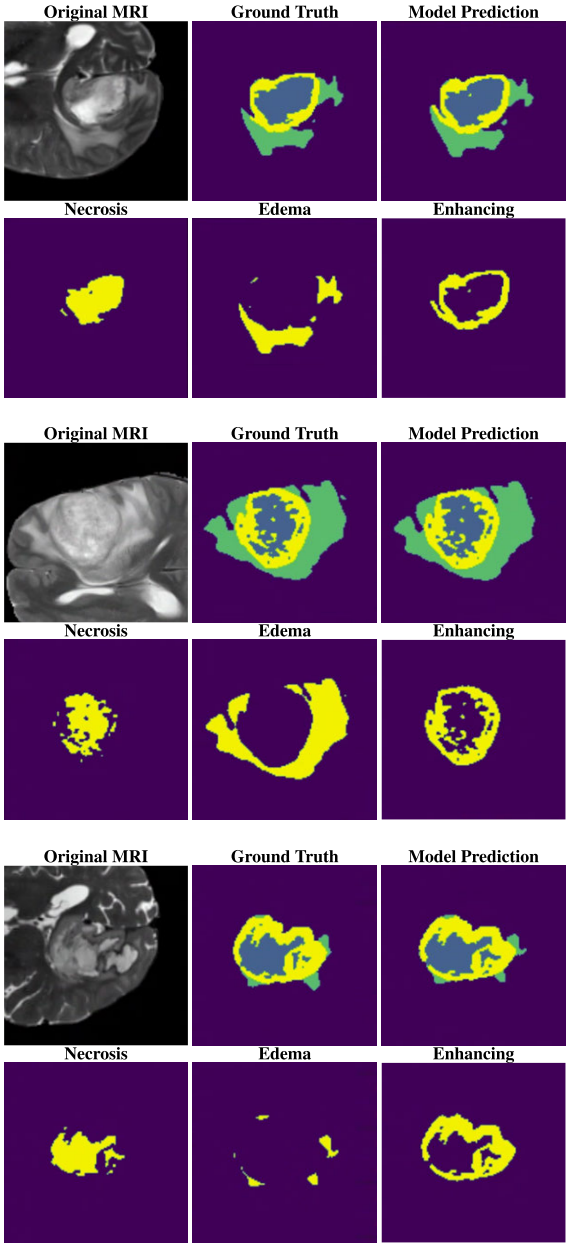
Patching Strategy	Without Attention Gates	With Attention Gates
Standard Random Cropping	0.84	0.86
Tumor-Centered Patching	0.88	**0.90**

Table 5demonstrates that UFTrans achieves superior dice score (Table 7) while being the most parameter-efficient model in the comparison (11M parameters vs. 31.2M for nnU-Net+ and 62.1M for Swin-UNETR). Crucially, UFTrans inference time (71 ms per volume) is 45× faster than nnU-Net+ and 82× faster than Swin-UNETR, with only 5 GB GPU memory required at inference, making it feasible for deployment on commodity clinical hardware. The 1–2% Dice improvements therefore come at a substantial *computational cost reduction*, not increase, establishing a favorable accuracy-efficiency trade-off for clinical adoption.

5) QUALITATIVE RESULTS

The qualitative results in Table 12 underscore the effectiveness of the proposed model in achieving accurate glioma segmentation across diverse cases. The segmentation outputs are visually compared against the ground truth, highlighting the model's ability to delineate tumor boundaries with high precision. The results show that the model successfully captures the tumor core (TC), enhancing tumor (ET), and whole tumor (WT) with clear structural consistency. The model demonstrates the ability to preserve fine tumor boundaries, particularly in cases with irregular growth patterns. The predictions closely align with the ground truth, showing minimal false positives and false negatives. The model effectively distinguishes between different tumor subregions, demonstrating strong contrast sensitivity and structural integrity in its segmentation. In samples where the tumor margins exhibit reduced clarity as a result of intensity fluctuations, the model continues to preserve a distinct differentiation between tumor and non-tumor regions, thereby indicating its strong feature extraction capabilities. Additionally, the model performs well in cases with heterogeneous tumor textures, where traditional CNN-based approaches often struggle. The segmentation outputs indicate that the model can differentiate between enhancing and non-enhancing tumor regions, that is crucial for clinical assessment.

TABLE 12. Qualitative results of our proposed model on BraTS 2021 dataset.



V. LIMITATIONS AND FUTURE WORK

A. LIMITATIONS

Despite promising results, several limitations should be noted. First, the proposed approach is evaluated only on the BraTS 2021 benchmark therefore the extent to which UFTrans generalizes to other tumor entities (e.g., meningioma and metastases), scanner vendors, acquisition protocols, and institutional distributions remains unclear and warrants further validation. A key limitation is evaluation on a single dataset (BraTS 2021). Multi-dataset validation is essential to establish robustness and is planned as immediate future work. Second, UFTrans is computationally demanding, requiring approximately 24 GB of GPU memory during training (Table 7), which may hinder reproducibility and limit adoption in resource-constrained settings. Third, class imbalance continues to affect segmentation performance: although focal loss alleviates this issue to some extent, accurately delineating small enhancing tumor components remains challenging and can reduce robustness in subtle or low-volume cases. Finally, the tumor-centered patching strategy assumes access to an approximate tumor location at inference time, which constrains fully automated deployment and may reduce applicability in scenarios where reliable localization is not readily available.

B. FUTURE WORK

Several directions can be pursued to strengthen and extend this work. A priority is to adopt more rigorous evaluation protocols, including stratified k-fold cross-validation with reporting of mean performance and standard deviations (and/or confidence intervals), to provide more reliable performance estimates and because BraTS test labels are not publicly released, we will submit UFTrans to the official BraTS validation/test server to obtain standardized blind evaluation results. In addition, validating UFTrans on newer BraTS editions (e.g., BraTS 2023) and other multi-institution brain tumor datasets would better demonstrate generalizability, particularly under domain shift. To improve deployment feasibility, future efforts will investigate model compression and acceleration techniques such as knowledge distillation, pruning, and mixed-precision inference to reduce memory footprint and latency. Methodologically, integrating frequency-domain representations may enhance

discrimination of heterogeneous tumor textures and complex patterns, while exploring multi-scale fusion across multiple encoder stages (rather than only at the bottleneck) could better capture both global context and fine-grained boundaries. Finally, submitting the model to the official BraTS validation servers would enable standardized benchmarking against other methods, and incorporating interpretability tools such as Grad-CAM or attention rollout could improve transparency and clinical trust.

VI. CONCLUSION

This study introduces a hybrid model that integrates local details from the 3D attention U-Net encoder with the long-range dependencies of the 3D vision transformer. Additionally, we incorporate an attention block in the 3D U-Net architecture that aids the model's convergence within fewer epochs by reducing the weights assigned to irrelevant features during training. The early fusion of global features from the 3D vision transformer with the 3D attention U-Net encoder's local details using Hadamard product, enhanced by an attention mechanism marks a significant advancement in brain tumor segmentation. Comprehensive experiments performed on the BraTS 2021 dataset demonstrate that our methodology surpasses contemporary state-of-the-art techniques, achieving Dice scores of 0.90 (WT), 0.91 (ET), and 0.89 (TC). Furthermore, we plan to extend our model in future work by integrating frequency-based features into the feature extraction process to identify more complex patterns within tumor regions.

DATA AVAILABILITY STATEMENT

The data supporting the study's findings is freely accessible at the BraTS 2021 Challenge repository on Synapse: https://www.synapse.org/#!Synapse:syn25829067/wiki/610863

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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